# Function-Space Decoupled Diffusion for Forward and Inverse Modeling in Carbon Capture and Storage

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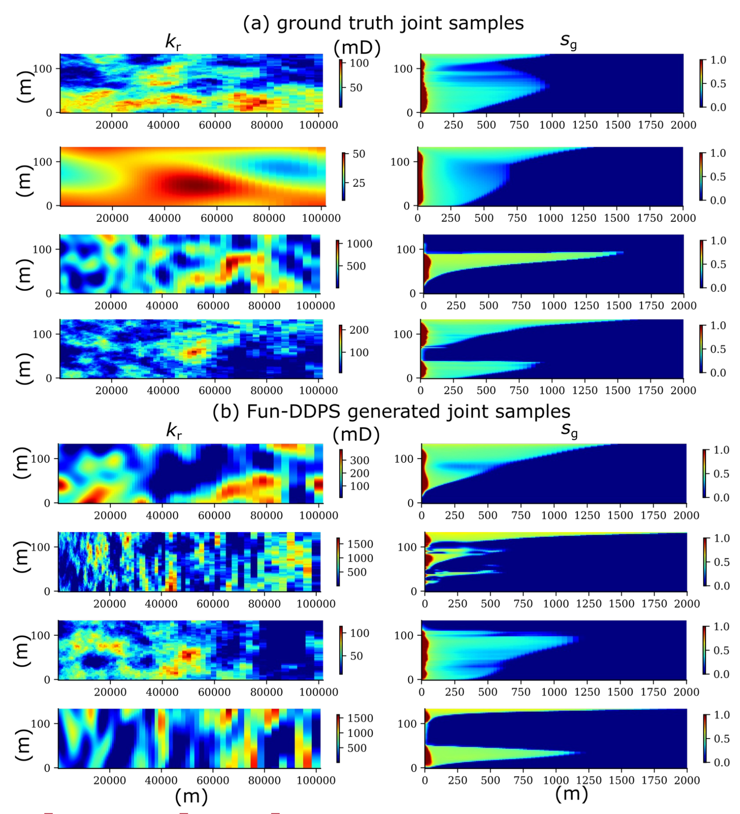
## Background

Carbon Capture and Storage (CCS) requires accurate subsurface flow modeling. Two tasks are central: forward prediction of CO2 plume migration and inverse characterization of geology from sparse monitoring data.

Both tasks face a fundamental bottleneck: high-dimensional, non-Gaussian geological parameters and expensive multiphase-flow solvers.




## Limitations of Existing Diffusion-based Solvers

Diffusion models can learn priors for scientific inverse problems. However, existing joint-state architectures, learn p(m, s) together and suffer from:

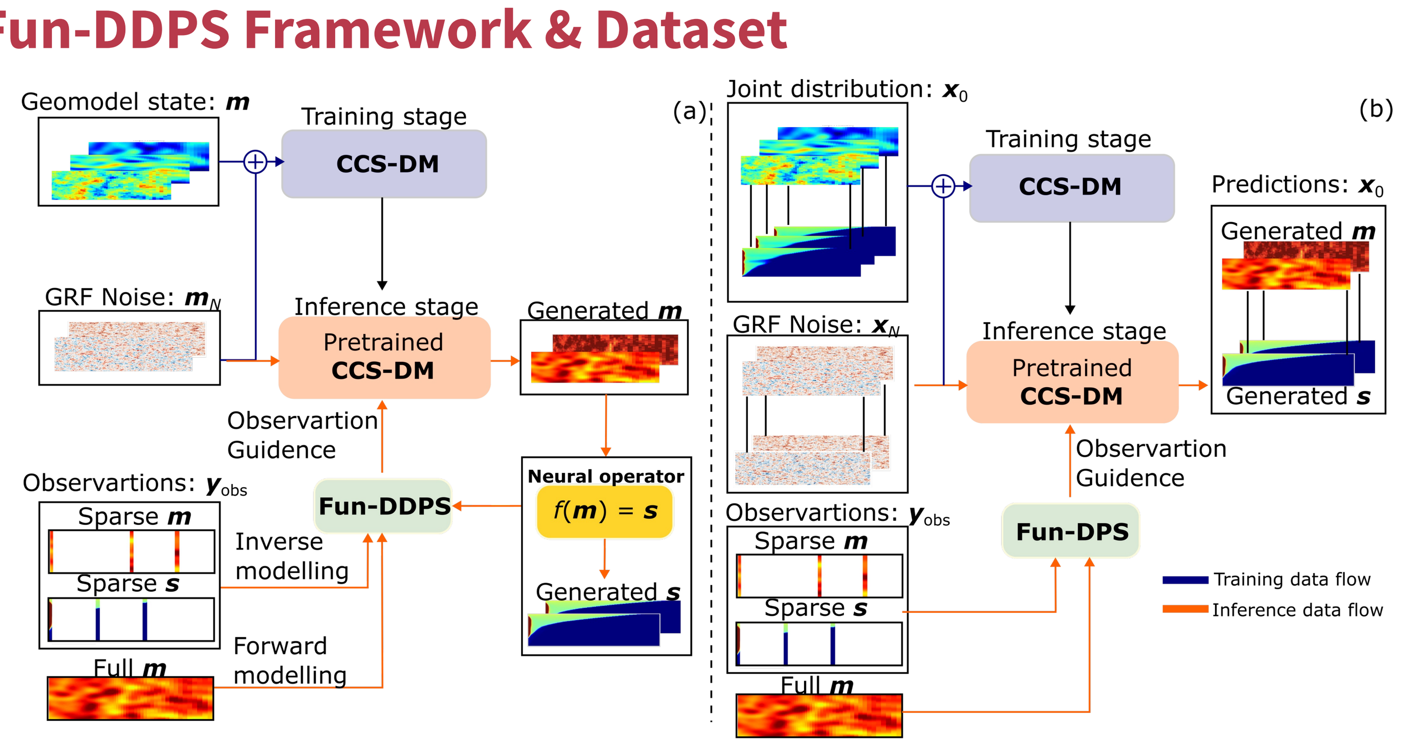
- Guidance attenuation under sparse observations — gradients vanish when training data is limited.
- High-frequency artifacts in posterior samples.
- Physical inconsistency — models learn statistical correlations rather than explicit physical laws.

We address these limits by decoupling the geological prior from the flow physics.

## Key Contributions

- Fun-DDPS framework: decouples the function-space diffusion prior p(m) from flow physics via a differentiable neural-operator surrogate $\mathcal{L}_\phi$ for likelihood guidance.
- Robust forward modeling: 7.7% rel. error at 25% coverage vs 86.9% for a deterministic surrogate — **an 11× improvement.**
- First rigorous validation of diffusion inverse solvers against asymptotically exact Rejection Sampling: JS < 0.06, **with 4× fewer function evaluations.**

## Fun-DDPS Framework & Dataset

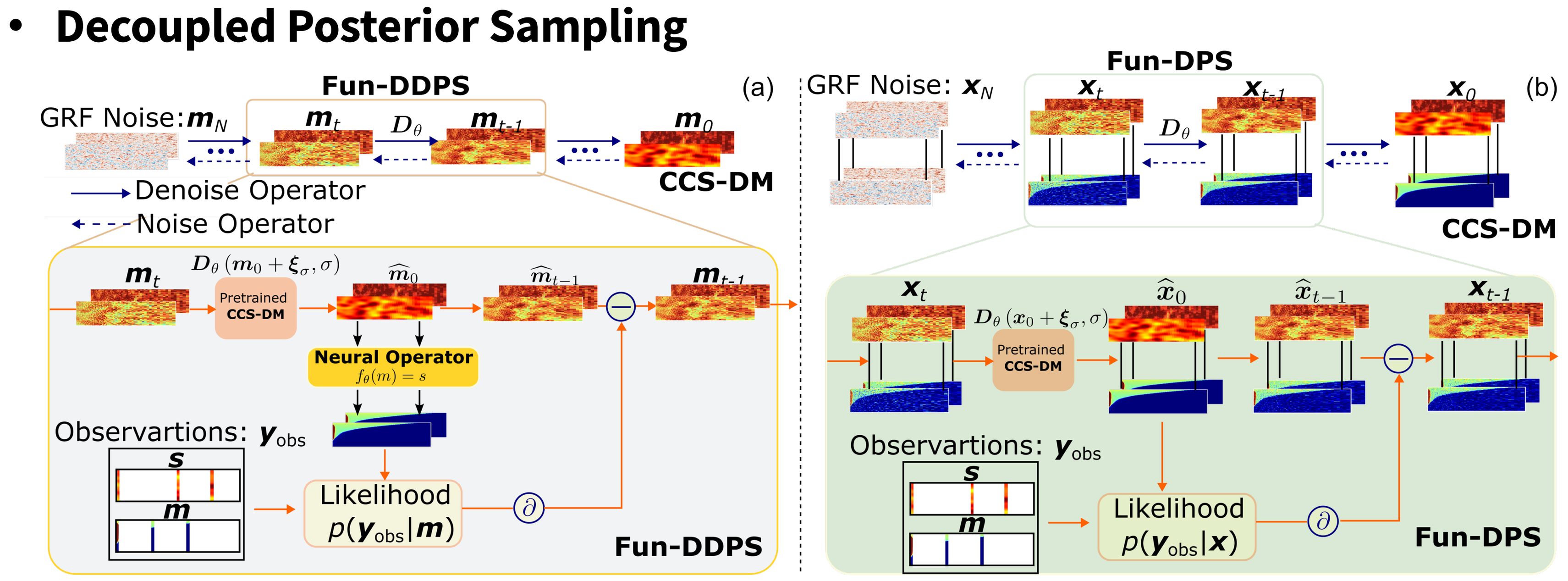



(a) **Fun-DDPS** (decoupled): train the diffusion model on the geomodel m alone, together with a separate neural-operator surrogate $\mathcal{L}_\phi$. Dynamics observations guide the prior through the surrogate gradient.
(b) **Fun-DPS** (joint-state): train on paired (m, s). Sparse dynamics observations remain localized, producing high-frequency artifacts.

## Reference

1. Lin et al., "Decoupled Diffusion Inverse Solver," arXiv, 2025.
2. Yao et al., "Guided diffusion for subsurface inverse problems," 2025. Chung et al., "Diffusion Posterior Sampling (DPS)," ICLR 2023.
3. Lim et al., "Score-based diffusion in function spaces," 2023. Liu et al., "Local Neural Operator (LNO)," 2024. Rahman et al., "U-shaped Neural Operator (U-NO)," 2022.

- **Decoupled Posterior Sampling**

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Conditional score decomposes as

$$\nabla_m \log p(m \mid y_{obs}) = \nabla_m \log p(m) + \nabla_m \log p(y_{obs} \mid m)$$

Forward task — direct geomodel likelihood

$$\nabla_m \log p(y_{geo} \mid \hat{m}_0) \approx -\zeta_{geo} \nabla_m \|M_{geo} \odot (\hat{m}_0 - y_{geo})\|^2_2$$

Inverse task — guidance through the surrogate

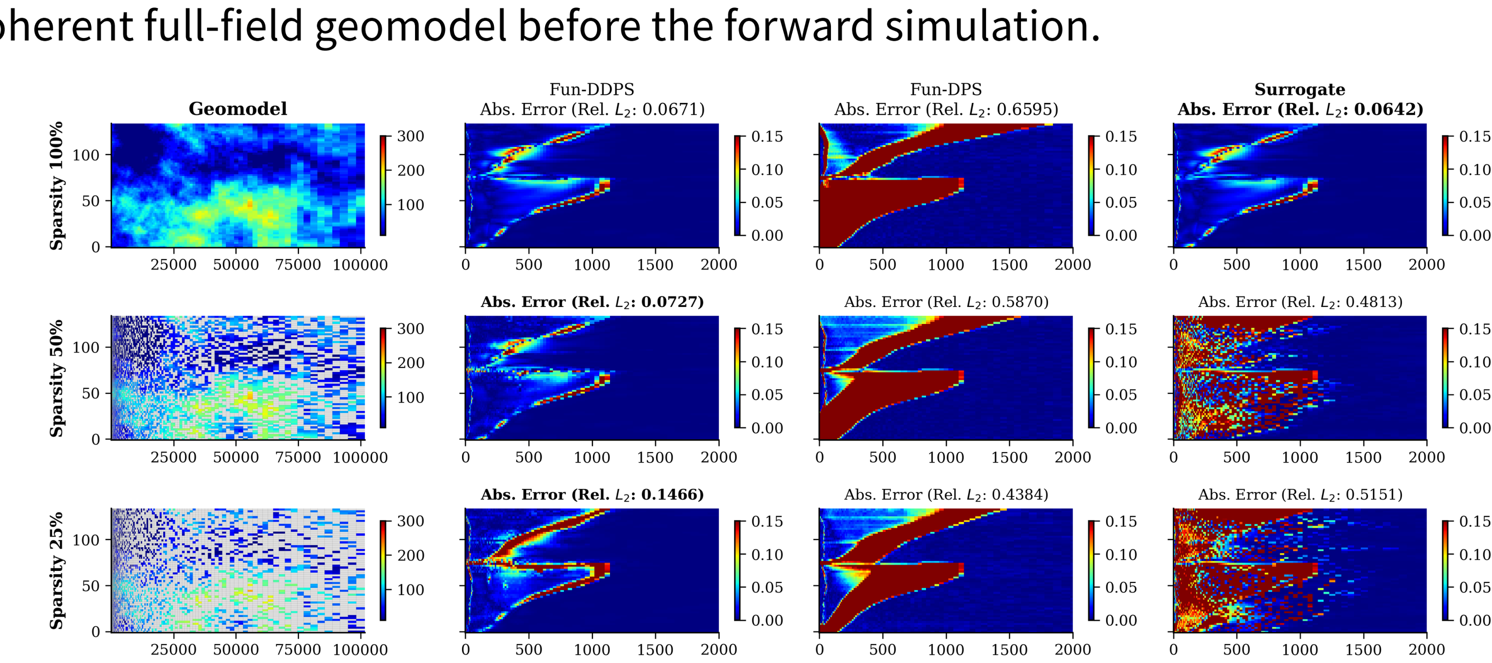
$$\nabla_m \log p(y_{dyn} \mid \hat{m}_0) \approx -\zeta_{dyn} \nabla_m \|M_{dyn} \odot (\mathcal{L}_\varphi(\hat{m}_0) - y_{dyn})\|^2_2$$

The surrogate's global receptive field turns sparse solution-space constraints into dense parameter-space guidance, avoiding the guidance attenuation of joint-state models.

- **CCS Dataset & Architectures**
  - Dataset. 12,000 training + 1,500 test pairs simulated with ECLIPSE (e300). Heterogeneous permeability m drawn via SGeMS; saturation s after 30 years of $CO_2$ injection in a radially symmetric deep saline aquifer.
  - Surrogate $\mathcal{L}_\phi$ — Local Neural Operator (LNO). Fourier spectral layers resolve global pressure; localised DISCO kernels capture shock-like saturation fronts without ringing artifacts.
  - Diffusion backbone — U-NO. A multi-scale U-shaped neural operator enforces discretisation invariance. Fun-DDPS trains on geomodels only; Fun-DPS trains on paired (m, s).

## Forward Problem — Sparse Geomodel Observation

Predict saturation ***s*** from partially observed geomodel m at 100%, 50%, and 25% coverage. Deterministic surrogates fail catastrophically — zero-filled inputs leave the training distribution — whereas Fun-DDPS uses the generative prior to reconstruct a coherent full-field geomodel before the forward simulation.



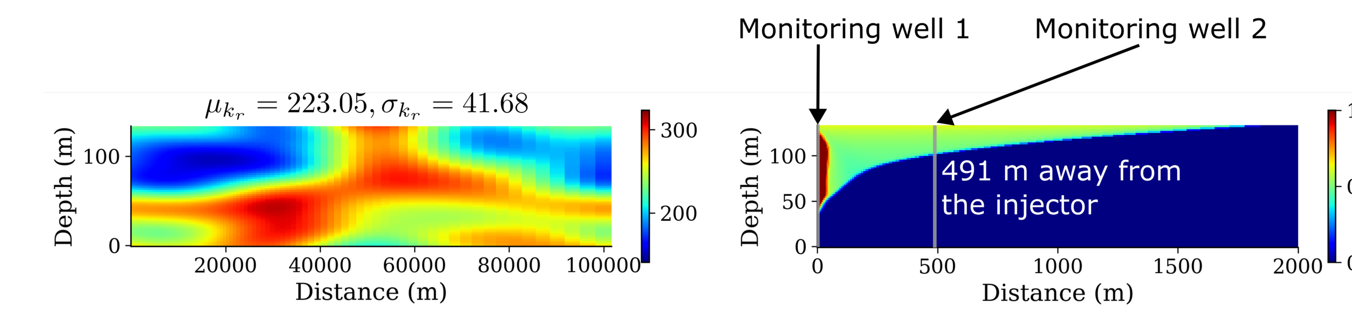

Saturation prediction error across methods at 100 / 50 / 25% observation coverage (rows). Fun-DDPS (col 2) maintains a tight error band; the deterministic surrogate (col 4) degrades sharply.

| Obs. coverage | Fun-DDPS | Fun-DPS | Surrogate |
|---|---|---|---|
| 100% | 0.046 ± 0.058 | 0.418 ± 0.366 | 0.044 ± 0.057 |
| 50% | 0.054 ± 0.069 | 0.370 ± 0.357 | 0.850 ± 0.263 |
| 25% | 0.077 ± 0.079 | 0.336 ± 0.349 | 0.869 ± 0.258 |

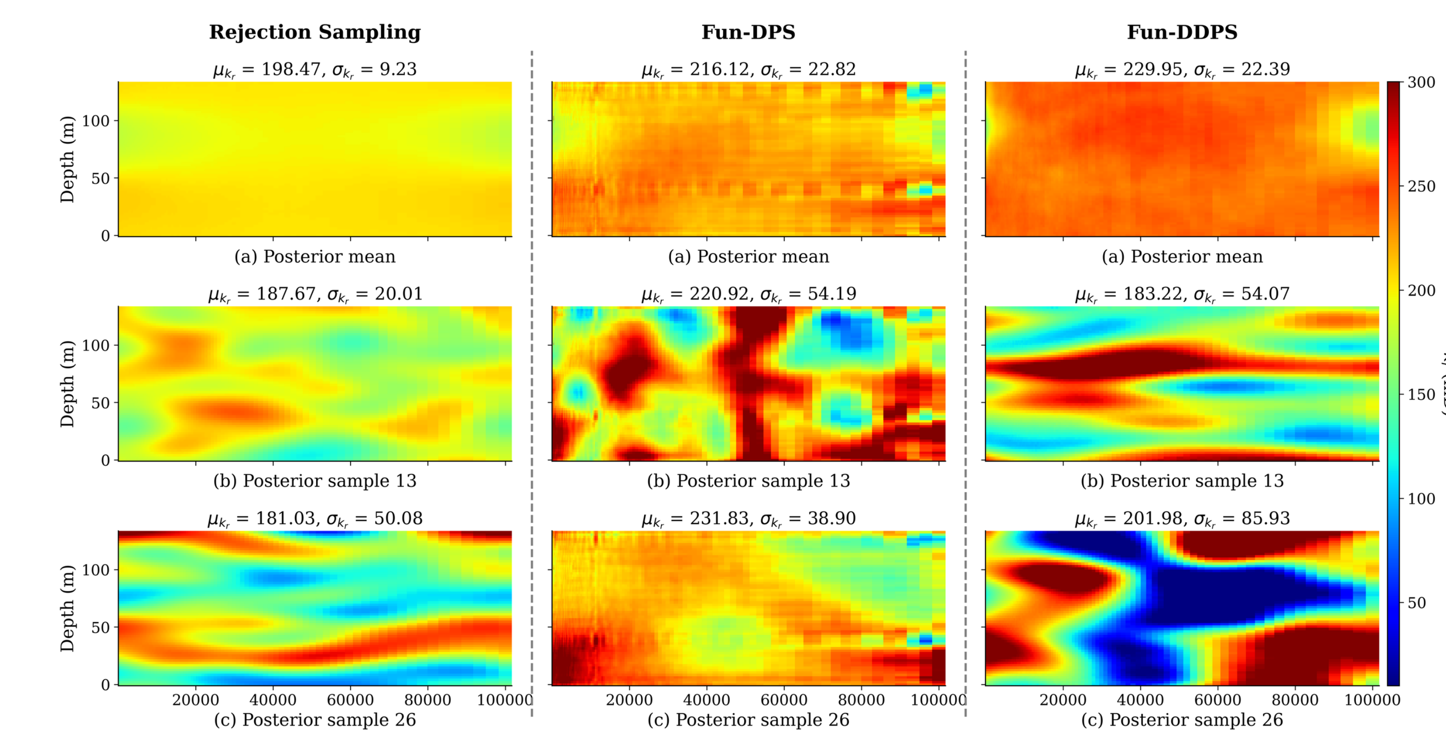
- Relative $L^2$ error (mean ± std). Fun-DDPS stays robust under sparsity; the deterministic surrogate collapses to 86.9% error at 25%.

## Inverse Problem — < 1% Dynamics Observation

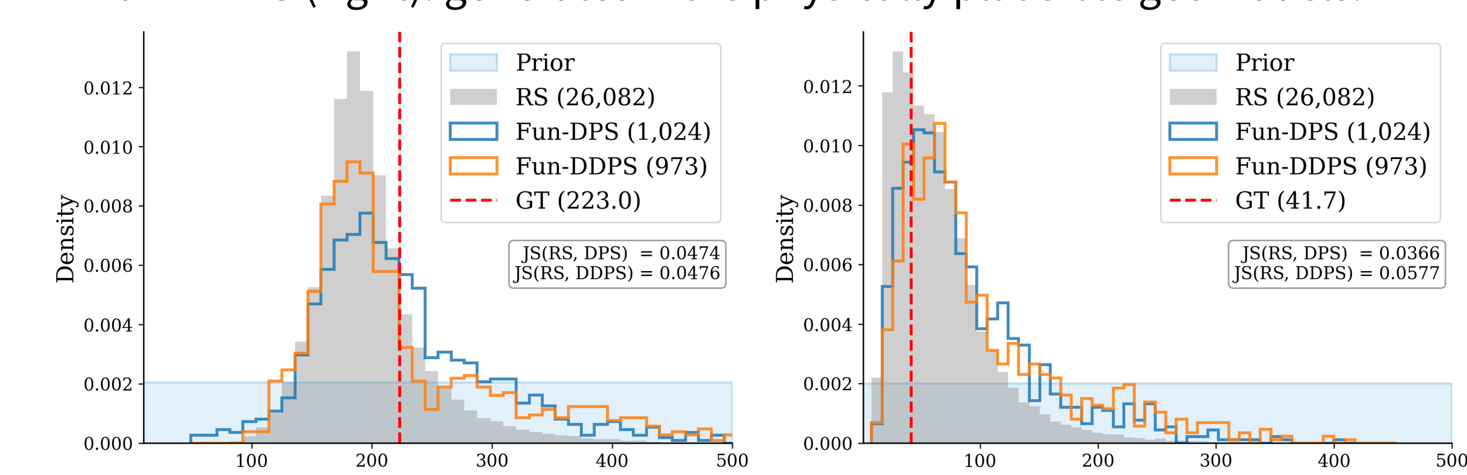
Two monitoring wells provide only 128 saturation data points (< 1% of the spatial domain, $\sigma_{obs} = 0.04$). Reference posterior via Rejection Sampling (26,082 accepted from 2M prior draws).




- **Posterior Geomodel Comparison**




- **RS** (left): ground-truth. **Fun-DPS** (center): captures posterior structure but suffers from high-frequency artifacts in the grainy posterior mean. **Fun-DDPS** (right): generates more physically plausible geomodels.




- **Fun-DPS** (JS=0.047) is statistically precise but produces a grainy posterior mean with high-frequency artifacts. **Fun-DDPS** (JS=0.051) yields geologically coherent fields matching the RS reference.

## Conclusion & Future work

- Fun-DDPS decouples the geological prior from flow physics, addressing extreme sparsity and computational cost in subsurface data assimilation.
- 11× improvement in forward modeling under 25% data coverage vs a deterministic surrogate.
- First rigorous validation of diffusion-based inverse solvers against Rejection Sampling posteriors.
- Physically consistent posterior samples with 4× efficiency gain over RS.
- Future work: extend to full spatiotemporal trajectories (4D seismic, pressure time series).

## Collaboration

Stanford University  Caltech  EarthFlowAI